

Technical Document #317: Database Systems - Comprehensive Overview

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Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

Database connection pooling reuses database connections to reduce overhead. It improves performance for applications with many concurrent users.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Key Takeaways:

- Database caching stores frequently accessed data in memory.
- NoSQL databases handle unstructured and semi-structured data.
- Database connection pooling reuses database connections to reduce overhead.